UU-B-00342C(GSA-FSS)
June 22, 1970
SUPERSEDING
Int. Fed. Spec. UU-B-00342b(GSA-FSS)
June 15, 1967, and
INTERIM REVISION OF
Fed. Spec. UU-B-342a
September 7, 1965

INTERIM FEDERAL SPECIFICATION

BINDER, LOOSE-LEAF, RING: ENGINEER'S FIELD BOOK

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification UU-B-342a dated September 7, 1965.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for closed-back (nonseparable), side-opening six-ring binders in two sizes, for firm retention of loose-leaf papers when closed but which can be readily released by the use of an opening mechanism(s) included in the binder. The smaller-size binder shall be furnished with or without embankment tables, as specified (see 6.2).

1.2 Classification.

1.2.1 Type and grade. The binders shall be of one type and one grade.

1.2.2 Size, number of rings, ring spacing, and capacity. Unless otherwise specified, size, number of rings, ring spacing, and capacity shall be as shown in table I. Sheet size shall be specified in the invitation for bid, contract, or order; and in the case of exceptions to table I, all information in the table shall be specified (see 6.2).

TABLE I. Sheet size, number of rings, ring spacing, and capacity

Size of sheet		No. of rings	Ring spacing center-to-center			Capacity
Ring side	Other side		Between rings	Between outside rings	Between sets of rings	
Inches	Inches		Inch	Inches	Inches	Inch
6-3/4	4	6	3/4	5	2	1/4
7-1/4	4-5/8	6	3/4	5-1/4	2-1/4	1/4

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-B-623 - Book Cover Board.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestics Shipment (Civilian Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Publication:

ASTM E 18 Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical Association of the Pulp and Paper Industry (TAPPI) Publications:

T410 - Weight Per Unit Area (Basis Weight or Substance) of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, New York 10017.)

National Classification Board:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attention: Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Book cover board. Book cover board shall conform to the requirements of UU-B-623, and shall be 0.098 inch in thickness. Thickness tolerance permitted shall be in accordance with the cited specification.

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3.1.2 Coated cloth.

3.1.2.1 Base cloth. The base cloth shall be bleached or unbleached cotton drill, and shall have not more than 5 percent nonfibrous materials. The yarns per inch shall be not less than 68 in the warp direction and 40 in the filling direction. The weight of the base cloth shall be not less than 5.3 ounces per square yard.

3.1.2.2 Coating compound. The coating compound shall be virgin plastic formulated from any of the following materials: pyroxylin, ethyl cellulose, cellulose acetate, polyvinyl chloride, or polyvinyl chloride-acetate. Plasticizer shall be adequate to provide flexibility for resistance to drying and cracking of the finished coated cloth when binders are in use. The weight of the coating shall be not less than 4.0 ounces per square yard. The coating shall be free from objectionable odor.

3.1.2.3 Finished coated cloth. The finished coated cloth shall consist of the base cloth and the coating compound. The base cloth shall be evenly and smoothly coated with coating compound on the face side, and shall be free from fisheyes, holes, cracks, or other defects. The finished cloth shall have a minimum breaking strength of 95 pounds in the warp direction and 69 pounds in the filling direction. The weight of the finished cloth shall be not less than 9.3 ounces per square yard. The resistance of the coated cloth to crocking shall have a rating of at least "fair" when tested in accordance with 4.1.1.

3.1.2.3.1 Color and finish. Unless otherwise specified, color of coated cloth shall be black (see 6.2), and shall have a Morocco grain, semi-bright finish at the option of the manufacturer.

3.1.3 Duck. The duck cloth shall be of cotton consisting of not less than single yarns in both the warp and filling direction. The yarns per inch shall be not less than 88 in the warp direction and 32 in the filling direction. The duck shall have a finished weight of not less than 6.0 ounces per square yard. The color shall be compatible with the color of the finished coated cloth.

3.1.4 Paper lining. Paper lining shall have a basis weight of not less than 60 pounds (24 by 36 inches - 500 sheets) when tested in accordance with 4.1.1. The color shall be compatible with the color of the finished coated cloth.

3.1.5 Steel. The steel shall be of good commercial quality, and all exposed steel surfaces shall have a polished finish.

3.1.5.1 Back plate. The back plate shall be of cold-rolled steel not less than 0.016 inch in thickness.

3.1.5.2 Ring plates (arch frames). The ring plates shall be of cold-rolled steel not less than 0.025 inch in thickness with a hardness of R30T 56.5-67.

3.1.5.3 Rings (arches). Rings shall be of steel having a hardness of R30T 56.5-67. Ring dimensions shall be in accordance with normal commercial practice.

3.1.5.4 Cover plate (spring or tension). The cover plate shall be of cold-rolled steel not less than 0.011 inch in thickness with a hardness of R15T 80-85.

3.1.6 Back plate covering. The back plate covering shall be flexible board or pasted board, not less than 0.020 inch in thickness.

3.2 Design, construction, and operational performance. Binders shall open on the side, and shall have two stiff covers and a closed, rounded back containing the ring binding mechanism. Each binder shall be designed to provide a secure place for filler sheets, and embankment tables, as applicable, in such a manner that one or more sheets can be removed or inserted in the binder without disturbing the remaining contents. Binders shall be designed so that sheets therein will present a flat writing surface for making entries when the binder is open. The binders shall be constructed so that rings can be closed manually and can be opened by means of a steel opening lever or other device at one or both ends of the binding mechanism. Rings shall fit snugly in closed position and shall be in proper alignment. Rings shall open and close satisfactorily and the binding mechanism shall operate smoothly without breakdown (see 4.3.3). The binder shall be constructed in a manner that no parts of the binder will mar smooth surfaces such as desks and tables.

3.2.1 Covers. Covers shall be of book cover board (3.1.1) bound with coated cloth (3.1.2) of one-piece (not spliced), securely glued to the entire area of contact with the back and covers. The coated cloth shall be neatly and securely pleated or turned at the corners of the covers on the index edges and the corners shall be rounded. The coated cloth shall be carried over onto the inside

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surfaces of the covers for at least 3/8 of an inch. Each inside cover shall have a paper lining (see 3.1.4) glued over the book cover board, the overlapped coated cloth binding material, and duck hinge material. The paper lining shall extend over the overlapped coated cloth in accordance with good commercial practice.

3.2.2 Binding mechanism. The binding mechanism of the binder shall consist of a steel back plate (3.1.5.1), two steel strip-type ring plates (3.1.5.2), and a steel cover plate (3.1.5.4). The rings (3.1.5.3) shall be securely attached to, or an integral part of, the ring plates. The combined width dimensions of the two ring plates (in position) shall closely approximate the internal width dimension of the cover plate. The length and width of the ring plates and cover plate shall be sufficient to ensure satisfactory operation of the mechanism. The width of the back plate for the larger sheet-size binder (table I) shall be not less than 5/8-inch and the smaller size, 1/2-inch. The opening and closing of the binding mechanism shall be controlled by the tension in the cover plate. The complete binding mechanism shall be attached to the back plate by at least two rivets with rivet heads concealed. The steel back plate shall have a securely attached covering of flexible board or pasted board (3.1.6). The covering shall be positioned between the back plate and coated cloth cover material, and shall be sufficiently longer and wider than the plate to protected the coated cloth from wear or cutting by the plate.

3.2.3 Hinges. Hinges shall consist of the coated cloth binding material and a strip of reinforcing duck (see 3.1.3). The duck shall be of one continuous piece covering the back plate, connecting the book cover board covers as required to provide a hinge, and extending for 3/4 ± 1/16 inch onto the inside of the covers. Sizing or other means of finishing to facilitate gluing, will be permitted.

3.3 Fly leaves. When specified (see 6.2), two black fly leaves, each not less than 0.015 inch thick, shall be furnished with each binder. Fly leaves, when furnished, shall be of pressboard, fiberboard, or plastic, with rounded corners. Fly leaves for the smaller sheet-size binder (table I) shall be 6-1/2 inches in length by 4 inches in width; for the larger sheet size, 7-3/16 inches in length and 4-1/2 inches in width. A plus of 1/16 inch or minus tolerance of 1/8 inch will be permitted in the length and width dimensions specified.

3.4 Embankment tables. When specified (see 6.2), the smaller sheet-size binder (Table I) shall be furnished with two tables, punched and inserted in the binder. One table shall consist of trigonometric formulae and the other of distances from side stakes for cross sectioning (see figures 1 and 2 respectively). The tables shall be imprinted, one on each side of an opaque, white vinyl sheet, laminated with a clear or transparent vinyl, measuring 6-1/4 inches in length by 4 inches in width and having a total thickness of at least 0.015 inch. A plus or minus tolerance of 1/8 inch will be permitted in the specified length and width dimensions. The printing shall be sharp and legible, without waviness or feathering, and of a permanent nature, i.e., will not fade and become illegible in normal use.

3.5 Identification of product marking. The manufacturer's name or trademark of such known character as to be readily identifiable, or both, shall be imprinted on the inside of one of the covers. When specified, the words "U.S. Government" shall be included in the marking (see 6.2). Additional marking in accordance with normal commercial practice will be acceptable.

3.6 Workmanship. The binders, including fly leaves and embankment tables, as applicable, shall be well finished, clean, and all edges shall be smooth. Metal parts of binders shall be free from burrs and rough or sharp edges. The coated cloth binding material at the four rounded corners of the binder covers shall be well-turned and shall be securely attached to the book cover board and paper lining. Binders, including fly leaves and embankment tables, as applicable, shall show no defects which might affect the serviceability of the items and shall be free from any defects which affect the appearance of the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Component testing of materials. In accordance with 4.1, the supplier is responsible for insuring that the following component materials were manufactured, examined, and tested in accordance with this specification or referenced subsidiary specifications or standards. Component testing shall represent materials received (or manufactured) by the supplier as a shipment at one time, of the same type and color, manufactured by the same source and essentially under the same conditions, and identified as used in the production of the end items. Unless otherwise specified, the sample size shall be the sample unit specified in the applicable test method, selected at random within the shipment.

Book Cover Board. Book cover board (3.1.1) shall be tested in accordance with the requirements of UU-B-623.

Base cloth, finished cloth, and duck. Base cloth, finished cloth, and duck shall be tested in accordance with the following test methods of Fed. Test Method Std. No. 191.

	Method
Base cloth (3.1.2.1):	
Yarns per inch	5050
Weight	5041
Finished coated cloth (3.1.2.3):	
Weight	5041
Breaking strength	5100
Crocking	5651
Duck (3.1.3):	
Weight	5041

Paper lining. The basis weight of paper lining (3.1.4) shall be determined in accordance with TAPPI Test Method T410.

Hardness and material thickness of steel. Hardness and material thickness of steel used in fabricating the binding mechanism and back plate shall be tested or inspected to determine conformance with the values specified in 3.1.5. Hardness of steel shall be determined in accordance with ASTM E 18.

4.1.1.1 Report of analysis. In lieu of using his own facilities, the supplier may submit a report of analysis of the component materials from a laboratory acceptable to the Government as evidence of compliance with the specification requirements. The supplier shall maintain records of component testing. These records shall be available to the authorized Government representative, and shall include: (1) the source, (2) date of receipt, (3) component material designation, (4) date of test(s), (5) test data, and (6) quantity of material represented by the test report. Each report of analysis must be within the period of the contract, or within 60 days prior to the commencement of the contract.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of all the binders presented for inspection at one time as defined in MIL-STD-105.

4.2.2 Sampling of end-item for visual defects. A random sample of binders shall be taken from each lot in accordance with MIL-STD-105, inspection level I. The acceptable quality level (AQL) shall not exceed 4.0 percent defective. The sample unit shall be one binder.

4.2.3 Sampling for preparation for delivery requirements. For examination of interior packages the sample unit shall be one shipping container fully prepared for delivery, and selected at random just prior to the closing operation. Closure defects shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. Sampling shall be in accordance with MIL-STD-105, inspection level I. The acceptable quality level (AQL) shall not exceed 4.0 percent defective.

4.2.4 Sampling for end-item operational performance test. A random sample of binders shall be taken from each lot in accordance with MIL-STD-105, inspection level S-2. The AQL shall not exceed 4.0 percent defective. The sample unit shall be one binder.

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4.3 Examination.

4.3.1 Visual and dimensional examination of end-item. Samples taken in accordance with 4.2.2 shall be examined for defects listed in table II. Any sample containing one or more defects shall be rejected.

TABLE II. Examination for visual and dimensional defects (end items)

Examine	Defects
Binder components and embankment tables, as applicable	Binder not size, capacity, color, and finish specified. Any dimension not as specified. Design and construction not as specified (see 3.2 through 3.2.3). Cover, hinges, and binding mechanism, as applicable, not as specified. Finish of exposed metal surfaces not as specified. Fly leaves (when required) not as specified; missing. Embankment tables (when required) not as specified; missing. Identification of product marking not as specified. Workmanship not as specified.

4.3.2 Examination of preparation for delivery. Samples taken in accordance with 4.2.3 shall be examined to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table III.

TABLE III. Examination of preparation for deliver

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size location, sequence, or method of application.
Materials (containers)	Not as specified. Any component missing; damaged in a manner to affect serviceability.
Closure of shipping containers	Not as specified.
Contents and weight, interior and exterior, as applicable	Quantity of binders not as specified. Gross or net weight exceeds limitation.
Workmanship	Inadequate application of components such as incomplete closure of container flaps; loose bands; inadequate sealing, strapping or stapling, as applicable. Bulged or distorted containers.
Packaging and packing containers	Not type, class, style, etc., specified, as applicable.

4.3.3 Examination for operational performance of binding mechanism. Samples taken in accordance with 4.2.4 shall be subject to the operational performance test indicated in table IV.

TABLE IV. Operational performance test of binding mechanism

Examine	Defects
Binding mechanism The binder shall be opened and closed in the normal manner of operation. Repeat this operation 20 consecutive times.	Opening lever or other opening device does not function properly, i.e., rings do properly. Rings do not open or close completely. Rings do not fit snugly in closed position; do not align properly.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), four binders of like description shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-655, or PPP-B-676.

5.1.2 Level B. Four binders of like description shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.3 Level C. The binders shall be packaged to afford protection against corrosion, deterioration and damage during shipment from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unit packages of binders in quantities as specified (see 6.2) shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed, waterproofed, and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. Ten unit packages of binders of like description shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The binders packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification.

5.3 Marking (see 6.2).

5.3.1 Civil agencies. In addition to markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

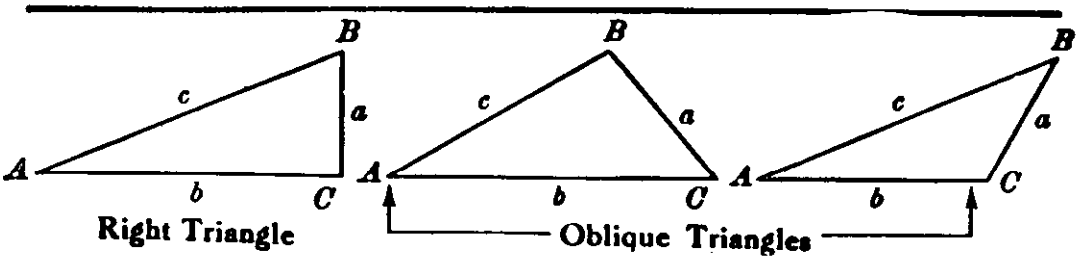
6.1 Intended use. The binders covered by this specification are intended for use in making notes pertaining to surveying in engineering field work.

6.2 Ordering data. Purchasers should select the preferred options offered herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Size (sheet size), and color of coated-cloth binding material if other than black is desired (see 1.2.2 and 3.1.2.3.1).
(c) Whether fly leaves are required (see 3.3).
(d) Whether embankment tables are required with the size 6-3/4 by 4-inch binder (see 3.4).
(e) Whether "U. S. Government" marking is required (see 3.5).
(f) Quantity of binders to a package if different than that specified for level A packaging (see 5.1.1).
(g) Quantity of packages to a shipping container for level A packing (see 5.2.1).
(h) Levels of packaging and packing required, and any special marking for shipment desired (see 5.1, 5.2, and 5.3).

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TRIGONOMETRIC FORMULÆ



Solution of Right Triangles

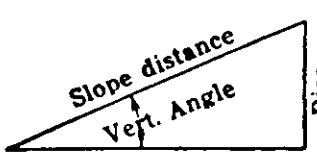
For Angle A. $\sin = \frac{a}{c}$, $\cos = \frac{b}{c}$, $\tan = \frac{a}{b}$, $\cot = \frac{b}{a}$, $\sec = \frac{c}{b}$, $\operatorname{cosec} = \frac{c}{a}$

Given	Required	
a, b	A, B, c	$\tan A = \frac{a}{b} = \cot B, c = \sqrt{a^2 + b^2} = a\sqrt{1 + \frac{b^2}{a^2}}$
a, c	A, B, b	$\sin A = \frac{a}{c} = \cos B, b = \sqrt{(c+a)(c-a)} = c\sqrt{1 - \frac{a^2}{c^2}}$
A, a	B, b, c	$B = 90° - A, b = a \cot A, c = \frac{a}{\sin A}$
A, b	B, a, c	$B = 90° - A, a = b \tan A, c = \frac{b}{\cos A}$
A, c	B, a, b	$B = 90° - A, a = c \sin A, b = c \cos A$,

Solution of Oblique Triangles

Given	Required	
A, B, a	b, c, C	$b = \frac{a \sin B}{\sin A}, C = 180° - (A + B), c = \frac{a \sin C}{\sin A}$
A, a, b	B, c, C	$\sin B = \frac{b \sin A}{a}, C = 180° - (A + B), c = \frac{a \sin C}{\sin A}$
a, b, C	A, B, c	$A + B = 180° - C, \tan \frac{1}{2}(A - B) = \frac{(a-b)\tan \frac{1}{2}(A+B)}{a+b}$, $c = \frac{a \sin C}{\sin A}$
a, b, c	A, B, C	$s = \frac{a+b+c}{2}, \sin \frac{1}{2}A = \sqrt{\frac{(s-b)(s-c)}{bc}}$, $\sin \frac{1}{2}B = \sqrt{\frac{(s-a)(s-c)}{ac}}, C = 180° - (A + B)$
a, b, c	Area	$s = \frac{a+b+c}{2}$, area $= \sqrt{s(s-a)(s-b)(s-c)}$
A, b, c	Area	area $= \frac{bc \sin A}{2}$
A, B, C, a	Area	area $= \frac{a^2 \sin B \sin C}{2 \sin A}$

REDUCTION TO HORIZONTAL



Horizontal distance = Slope distance multiplied by the cosine of the vertical angle. Thus: slope distance = 319.4 ft. Vert. angle = 5° 10′. From Table, Page IX. cos 5° 10′ = .9959. Horizontal distance = 319.4 × .9959 = 318.09 ft.

Horizontal distance also = Slope distance minus slope distance times (1 − cosine of vertical angle). With the same figures as in the preceding example, the following result is obtained. Cosine 5° 10′ = .9959. 1 − .9959 = .0041. 319.4 × .0041 = 1.31. 319.4 − 1.31 = 318.09 ft.

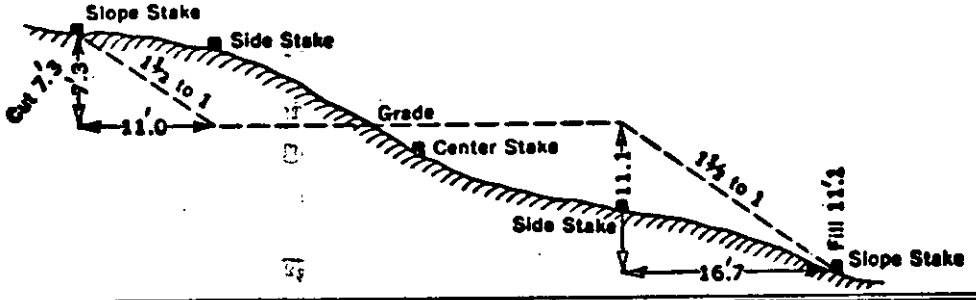
When the rise is known, the horizontal distance is approximately:—the slope distance less the square of the rise divided by twice the slope distance. Thus: rise = 14 ft., slope distance = 302.6 ft. Horizontal distance = $302.6 - \frac{14 \times 14}{2 \times 302.6} = 302.6 - 0.32 = 302.28$ ft.

Figure 1.

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DISTANCES FROM SIDE STAKES FOR CROSS-SECTIONING
Roadway of any Width. Side Slopes 1½ to 1.

In the figure below: opposite 7 under "Cut or Fill" and under .3 read 11.0, the distance out from the side stake at left. Also, opposite 11 under "Cut or Fill" and under .1 read 16.7, the distance out from the side stake at right.



Cut or Fill	0	.1	.2	.3	.4	.5	.6	.7	.8	.9	Cut or Fill
	Distance out from Side or Shoulder Stake										
0	0.0	0.2	0.3	0.5	0.6	0.8	0.9	1.1	1.2	1.4	0
1	1.5	1.7	1.8	2.0	2.1	2.3	2.4	2.6	2.7	2.9	1
2	3.0	3.2	3.3	3.5	3.6	3.8	3.9	4.1	4.2	4.4	2
3	4.5	4.7	4.8	5.0	5.1	5.3	5.4	5.6	5.7	5.9	3
4	6.0	6.2	6.3	6.5	6.6	6.8	6.9	7.1	7.2	7.4	4
5	7.5	7.7	7.8	8.0	8.1	8.3	8.4	8.6	8.7	8.9	5
6	9.0	9.2	9.3	9.5	9.6	9.8	9.9	10.1	10.2	10.4	6
7	10.5	10.7	10.8	11.0	11.1	11.3	11.4	11.6	11.7	11.9	7
8	12.0	12.2	12.3	12.5	12.6	12.8	12.9	13.1	13.2	13.4	8
9	13.5	13.7	13.8	14.0	14.1	14.3	14.4	14.6	14.7	14.9	9
10	15.0	15.2	15.3	15.5	15.6	15.8	15.9	16.1	16.2	16.4	10
11	16.5	16.7	16.8	17.0	17.1	17.3	17.4	17.6	17.7	17.9	11
12	18.0	18.2	18.3	18.5	18.6	18.8	18.9	19.1	19.2	19.4	12
13	19.5	19.7	19.8	20.0	20.1	20.3	20.4	20.6	20.7	20.9	13
14	21.0	21.2	21.3	21.5	21.6	21.8	21.9	22.1	22.2	22.4	14
15	22.5	22.7	22.8	23.0	23.1	23.3	23.4	23.6	23.7	23.9	15
16	24.0	24.2	24.3	24.5	24.6	24.8	24.9	25.1	25.2	25.4	16
17	25.5	25.7	25.8	26.0	26.1	26.3	26.4	26.6	26.7	26.9	17
18	27.0	27.2	27.3	27.5	27.6	27.8	27.9	28.1	28.2	28.4	18
19	28.5	28.7	28.8	29.0	29.1	29.3	29.4	29.6	29.7	29.9	19
20	30.0	30.2	30.3	30.5	30.6	30.8	30.9	31.1	31.2	31.4	20
21	31.5	31.7	31.8	32.0	32.1	32.3	32.4	32.6	32.7	32.9	21
22	33.0	33.2	33.3	33.5	33.6	33.8	33.9	34.1	34.2	34.4	22
23	34.5	34.7	34.8	35.0	35.1	35.3	35.4	35.6	35.7	35.9	23
24	36.0	36.2	36.3	36.5	36.6	36.8	36.9	37.1	37.2	37.4	24
25	37.5	37.7	37.8	38.0	38.1	38.3	38.4	38.6	38.7	38.9	25
26	39.0	39.2	39.3	39.5	39.6	39.8	39.9	40.1	40.2	40.4	26
27	40.5	40.7	40.8	41.0	41.1	41.3	41.4	41.6	41.7	41.9	27
28	42.0	42.2	42.3	42.5	42.6	42.8	42.9	43.1	43.2	43.4	28
29	43.5	43.7	43.8	44.0	44.1	44.3	44.4	44.6	44.7	44.9	29
30	45.0	45.2	45.3	45.5	45.6	45.8	45.9	46.1	46.2	46.4	30
31	46.5	46.7	46.8	47.0	47.1	47.3	47.4	47.6	47.7	47.9	31
32	48.0	48.2	48.3	48.5	48.6	48.8	48.9	49.1	49.2	49.4	32
33	49.5	49.7	49.8	50.0	50.1	50.3	50.4	50.6	50.7	50.9	33
34	51.0	51.2	51.3	51.5	51.6	51.8	51.9	52.1	52.2	52.4	34
35	52.5	52.7	52.8	53.0	53.1	53.3	53.4	53.6	53.7	53.9	35
36	54.0	54.2	54.3	54.5	54.6	54.8	54.9	55.1	55.2	55.4	36
37	55.5	55.7	55.8	56.0	56.1	56.3	56.4	56.6	56.7	56.9	37
38	57.0	57.2	57.3	57.5	57.6	57.8	57.9	58.1	58.2	58.4	38
39	58.5	58.7	58.8	59.0	59.1	59.3	59.4	59.6	59.7	59.9	39
40	60.0	60.2	60.3	60.5	60.6	60.8	60.9	61.1	61.2	61.4	40

Figure 2.

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to: Gen. Services Admin., FSS (FMSO), Washington, D.C. 20406.
NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

UU-B-00342C(GSA-FSS) BINDER, LOOSE-LEAF, RING: ENGINEER'S FIELD BOOK

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT. AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

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